Reusable component

DESIGN DOCUMENT + POST-DEPLOY RUNBOOK

FS Inspection Data Update Platform event publisher

Publishes FS_InspectionDataUpdate__e when a Field Services appointment is confirmed, scheduled, dispatched, unscheduled, or canceled. Assigned Resource changes stamp the inspector lookup and do not publish by themselves.

Org	AKLCDEVSB4
Status	Deployed
Deploy Id	0Af8r00000XThEiCAL
Tests	FS_SaStatusEventHandlerTest 19 / 19 passing
Event	FS_InspectionDataUpdate__e (existing; not overwritten)
Event type	InspectionStatusUpdate (single type)
Source API	61.0
Classification	Confidential

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1. Purpose

Salesforce publishes one High Volume platform event, `FS_InspectionDataUpdate__e`, so a subscriber application can keep inspection bookings in step with Field Service.

There is a single message type. Confirmation, schedule, reschedule, dispatch, unschedule, and cancel all use `EventType__c = InspectionStatusUpdate`. The JSON body is `FS_InspectionStatusPayloadDTO`. Properties with no value are omitted.

Travel, OnSite, and Completed status changes do not publish by themselves.

Manual inspections are stamped `FS_IsManualInspection__c = true` on the Service Appointment. That insert publishes one event, and the payload includes isManualInspection. Other creates do not publish. The same checkbox exists on Work Order and Work Order Line Item so CRM can stamp the booking hierarchy.

2. Goals and non-goals

Goals

- Publish a manual inspection on insert. Other creates do not publish. Publish when FS_Is_Confirmed__c changes to true on update.
- Publish on update when the appointment is dispatched, canceled, fully unscheduled, or scheduled with an inspector and both schedule times that just changed.
- Stamp ServiceAppointment.FS_Assigned_Inspector__c from Assigned Resource insert, resource change, and delete.
- Always set EventType__c to InspectionStatusUpdate.
- Publish manual inspections and include isManualInspection. Skip any record type other than Field_Services.
- Leave Alert Center parent-lookup fill and FSL managed triggers in place.

Non-goals

- Do not create the platform event. It already exists in the org.
- Do not publish from the Assigned Resource trigger. It only writes the inspector lookup.
- Do not publish on Service Appointment insert.
- Do not stamp FS_IsManualInspection__c from this package.
- Do not subscribe to the event in this package.

3. Architecture

Scheduling is two saves. Field Service writes the Service Appointment (status and times) and the Assigned Resource separately. The appointment trigger waits until the inspector lookup and both times are populated, so the subscriber receives one schedule event that already contains both.

```
ServiceAppointment after insert and after update
FS_SaStatusEventTrigger → MetadataTriggerHandler
Trigger_Action.FS_SaStatusEventAfterInsert
Trigger_Action.FS_SaStatusEventAfterUpdate
FS_SaStatusEventHandler.resolveEventType
query, map, EventBus.publish

AssignedResource after insert, update, and delete
FS_AssignedResourceInspectorTrigger → MetadataTriggerHandler
Trigger_Action.FS_AssignedInspectorAfterInsert / AfterUpdate / AfterDelete
FS_AssignedResourceInspectorHandler writes FS_Assigned_Inspector__c
That Service Appointment update is what publishes.
```

Layer	Member	Role
Trigger	FS_SaStatusEventTrigger	SA after insert and after update
Trigger	FS_AssignedResourceInspectorTrigger	AR after insert, update, delete
Handler	FS_SaStatusEventHandler	Publish decision, query, publish

Layer	Member	Role
Handler	FS_AssignedResourceInspectorHandler	Stamps the inspector lookup
Mapper	FS_InspectionStatusEventMapper	Header and JSON payload
DTO	FS_InspectionStatusPayloadDTO	JSON property names
Settings	FS_InspectionStatusEventSettings	Default_Setting CMDT
Field	FS_Is_Confirmed__c	Update publishes when it becomes true
Field	FS_Assigned_Inspector__c	Lookup to ServiceResource. Confidential
Perm set	FS_InspectionDataUpdate_Event_Publisher	Publisher users only

Each Trigger Action record has one context. Insert, update, and delete are separate records that point at the same Apex class. Parent Work Order lookups are filled by Alert Center, not by this package.

4. Publication rules

resolveEventType is the only publish decision. The first matching rule wins, so one save produces at most one event. These gates apply first: Default Setting Is Active is true (missing CMDT defaults to active); record type is Field_Services. Manual inspections are included, and the payload sets isManualInspection. FS_Is_Confirmed__c is not one of these gates.

Create

A manual inspection publishes on insert. Any other insert does not publish, even when FS_Is_Confirmed__c is already true. Confirmation of those appointments is a later update.

Update — first match wins

Order	Publish when
1	Status changed to Dispatched. Inspector and times are not required.
2	Status changed to Canceled. Inspector and times are not required. The org still requires FS_Cancelled_Reason__c.
3	FS_Is_Confirmed__c changed to true.
4	Status is None, the inspector lookup is blank, and status or the inspector just changed.
5	Status is Scheduled, the inspector and both times are populated, and status, the inspector, or the times just changed.

A drag onto another resource saves the appointment and the Assigned Resource separately. The handler queues that appointment once and publishes after the transaction commits, so the subscriber gets one event with the final inspector and times. A later save that leaves the appointment unpublished, such as status None while the inspector is still set, sends nothing. Rule 5 also publishes when status flips to Scheduled after the inspector and both times are already stored. It is one event for a new inspector, new times, or both together. A Scheduled save does not publish while the inspector lookup is blank. The Assigned Resource stamp writes the lookup, and that later appointment update publishes the inspector and both times together. If the inspector is stamped first, the update that fills both times publishes instead.

A change to None while the inspector is still populated does not publish. Clearing the inspector while status is still Scheduled does not publish. The unschedule event is published once both are true.

Does not publish

Situation	Result
Not Field_Services, or CMDT inactive	Skip
Non-manual Service Appointment insert	Skip
Scheduled while inspector or either time is blank	Skip
Arrival window, due date, actuals, or FS date stamps only	Skip

Situation	Result
Travel, OnSite, or Completed with no other matching change	Skip
None while the inspector lookup is still populated	Skip
Inspector cleared while status is still Scheduled	Skip
Already Canceled or Dispatched, status unchanged	Skip
Assigned Resource travel-time-only update	Lookup unchanged

Field Services statuses: None, Scheduled, Dispatched, Travel, OnSite, Completed, Canceled. FS_Inspection_Status__c, including OPEN, is a payload field and is not a publish trigger. Unscheduled_Status__c names the unscheduled status (None). It is not a separate allowlist.

5. Assigned inspector stamp

FS_AssignedResourceInspectorHandler maintains ServiceAppointment.FS_Assigned_Inspector__c (lookup to ServiceResource, relationship FS_Assigned_Inspections). It does not call EventBus.publish.

Assigned Resource DML	Lookup result
Insert with a Service Resource	Set the lookup. Last row in the batch wins for one appointment.
ServiceResourceId changes	Set the lookup to the new Service Resource.
Other field changes, such as travel time	No Service Appointment write.
Blank appointment or blank Service Resource	Ignored.
Delete, other resources remain	Keep the earliest remaining row by CreatedDate.
Delete, none remain	Clear the lookup.

- Only Field Services appointments are updated.
- The write is skipped when the lookup already matches, so the appointment trigger does not fire again.
- stampInProgress stops the appointment update from re-entering the stamp in the same transaction.
- Interactive users need update access on FS_Assigned_Inspector__c. CloudIntegrationUser and AutomatedProcess stamp in system mode.

6. Confirmation and manual inspections

FS_Is_Confirmed__c (Is Confirmed) defaults to false. The booking process sets it on an update. That update publishes when the flag changes to true. Insert does not publish. Dispatch, cancel, unschedule, and schedule updates still publish if the flag is left false. The publisher permission set grants read, not edit.

FS_IsManualInspection__c is a checkbox on Work Order, Work Order Line Item, and Service Appointment. Inserting a Service Appointment with the flag true publishes one event. The payload sets isManualInspection. CRM stamps all three on a manual booking. The publisher set grants read and edit.

7. Platform event and payload

FS_InspectionDataUpdate__e is High Volume and Publish After Commit. Header EventType__c, BookingReferenceNumber__c, InspectionId__c (FS_ExternalRefId__c), WorkOrderId__c, WorkOrderLineItemId__c, ServiceAppointmentId__c, and Payload__c. InspectionId__c matches payload sfsInspectionId.

Payload key	Source
status	ServiceAppointment.Status
bookingReferenceNumber	WorkOrder.FS_Booking_Reference__c
bookingNotes	WorkOrder.Description
crmInspectionId	FS_CRMInspection_Id__c

Payload key	Source
sfsInspectionId	FS_ExternalRefId__c
bcoNumber / propertyId / propertyAddress	BCO_Detail__c number, property id, address
contactInformation	First Inspection_Site_Contact__c name, phone, email
inspectionType	WorkType.Name text before the first underscore
isConfirmed	FS_Is_Confirmed__c
isManualInspection	FS_IsManualInspection__c
assignedInspectorId / Name	FS_Assigned_Inspector__c and its Name
cancelledReason / cancelledDate	FS_Cancelled_Reason__c / FS_Cancelled_Date__c
schedStartTime / schedEndTime	SchedStartTime / SchedEndTime
assignedCrewMembers[]	AssignedResource rows. Omitted when status is None
lastModifiedBy.name / username	LastModifiedBy display name and login

JSON.serialize omits nulls. On an unschedule event, status is None and assignedInspectorId and assignedCrewMembers are absent. On a cancel event, read status and cancelledReason. There is no separate cancellation event type. Travel time is minutes. estimatedTravelTimeSeconds is minutes times 60.

8. Classes and configuration

Method	Responsibility
afterInsert / afterUpdate	Cast TAF lists and call handleTrigger
handleTrigger	Active setting, Field Services filter, resolve, query, publish
resolveEventType	No create. Confirm, Dispatched, Canceled, unschedule, schedule
queryAppointments	WITH SECURITY_ENFORCED, then system mode if FLS blocks a field
stamp saveInspectors	Field Services lookup write, skipped when unchanged

Background users CloudIntegrationUser and AutomatedProcess query and publish in system mode. An interactive user who can read Service Appointment but lacks FLS on a payload field is retried in system mode so scheduling is not blocked. Org after-create automation requires FS_BCO_Detail__c.

Default Setting field	Value	Purpose
Is_Active__c	true	Master publish switch
Event_Type__c	InspectionStatusUpdate	Header and payload eventType
Unscheduled_Status__c	None	Unschedule rule and empty crew list
Scope_To_Field_Services_Record_Type__c	true	Apex always requires Field_Services

9. Publisher and subscriber permissions

FS_InspectionDataUpdate_Event_Publisher is the publisher set. This package does not deploy a subscriber permission set.

Publisher set

- Apex: FS_SaStatusEventHandler, FS_AssignedResourceInspectorHandler, mapper, settings, DTO.
- Create and Read on FS_InspectionDataUpdate__e. Read on BCO_Detail__c.
- ServiceAppointment.Field_Services record type visibility.
- Read and edit FS_Assigned_Inspector__c, and FS_IsManualInspection__c on Work Order, Work Order Line Item, and Service Appointment.

- Read, not edit: FS_Is_Confirmed__c, Status, and WorkOrder.FS_Booking_Reference__c, plus the other payload fields on the set.
- Required platform-event fields are available with object Create and Read. Do not add field permissions that the deploy rejects.

Assign the publisher set to

- Dispatchers, and field or mobile users who dispatch or cancel.
- Booking users whose insert of a confirmed appointment runs as that user.
- Integration users that update Service Appointments or Assigned Resources and must cause a publish.
- Do not assign it to CloudIntegrationUser or AutomatedProcess. Those users already run in system mode.

The process that sets Is Confirmed needs edit on FS_Is_Confirmed__c from its own profile or permission set.

Subscriber application user

- Grant Read on FS_InspectionDataUpdate__e. Do not grant Create unless that user also publishes.
- Do not assign the publisher set only so the application can listen.
- A Flow, Apex subscriber, or streaming client running as a named user needs Read on the event.
- Store subscriber credentials in the subscriber application. This package has no endpoints or tokens.
- Consume FS_InspectionDataUpdate__e only. Branch on EventType__c = InspectionStatusUpdate and payload status.
- Map assignedInspectorId, assignedInspectorName, sfsInspectionId, crmInspectionId, cancelledReason, and isConfirmed.
- A missing JSON key means null. Unschedule has status None and no inspector. Cancel has status Canceled.
- Give the subscriber no edit rights on Service Appointment or Assigned Resource unless that user may change bookings.

10. Tests

Deploy 0Af8r00000XThEiCAL ran FS_SaStatusEventHandlerTest: 19 passed, 0 failed. Tests call the handlers directly and still run Field Service DML.

Test	Expectation
testCreateDoesNotPublish	Insert does not publish
testConfirmedUpdatePublishesEvent	Is Confirmed changing to true publishes
testStatusBecomesScheduledPublishesEvent	Status flip to Scheduled publishes when inspector and times exist
testNewInspectorAndNewTimesPublishesEvent	One event with inspector and both times
testScheduleChangeOnScheduledPublishesEvent	Same inspector, new times, one event
testInspectorChangePublishesEvent	Inspector change on Scheduled publishes
testScheduledWithoutTimesDoesNotPublish	Scheduled without both times does not publish
testDispatchedPublishesEvent	Dispatched publishes
testCanceledStatusPublishesEvent	Canceled publishes, with cancelledReason
testUnscheduleDoesNotPublish	None with an inspector still set does not publish
testUnschedulePublishesWhenInspectorCleared	None and a blank inspector publishes
testAssignedResourceStampsInspector	Insert, reassignment, and delete maintain the lookup
testManualInspectionPublishesEvent	A scheduled manual inspection publishes, isManualInspection true
testInactiveSettingDoesNotPublish	Inactive CMDT publishes nothing

11. Post-deploy steps

Metadata deploy does not assign permission sets, add fields to layouts, set Is Confirmed, stamp CRM backfills, or connect the subscriber.

11.1 Publisher permission set

- Setup, Users, Permission Sets, FS_InspectionDataUpdate_Event_Publisher, Manage Assignments.
- Assign the users in section 9. Leave CloudIntegrationUser and AutomatedProcess unassigned.

11.2 Is Confirmed edit

Grant the booking or confirmation process edit on ServiceAppointment.FS_Is_Confirmed__c. Insert stays silent. The update that sets the flag to true publishes.

11.3 Subscriber application user

- Identify the integration user the subscriber runs as.
- Grant Read on FS_InspectionDataUpdate__e only.
- Assign the publisher set only if that same user also updates appointments.
- Point the subscriber at this one event and at EventType__c = InspectionStatusUpdate.
- Confirm it maps the payload keys in section 7.

11.4 Page layouts

- Work Order and Work Order Line Item CRM layouts: Is Manual Inspection.
- Field Services Service Appointment layout: Is Manual Inspection, Is Confirmed, and Assigned Inspector.
- Prefer Assigned Inspector read-only. The Assigned Resource trigger overwrites a manual edit on the next assign or unassign.

11.5 Default Setting and Trigger Actions

- Default Setting: Is Active true, Event Type InspectionStatusUpdate, Unscheduled Status None.
- Service Appointment actions FS_SaStatusEventAfterInsert and FS_SaStatusEventAfterUpdate, class FS_SaStatusEventHandler, one context each, not bypassed.
- Assigned Resource actions FS_AssignedInspectorAfterInsert, AfterUpdate, and AfterDelete, class FS_AssignedResourceInspectorHandler, not bypassed.
- Only one unpackaged trigger on each object calls MetadataTriggerHandler. Leave FSL TR001_Service, TR021_AssignedResource, and Alert Center FS_ServiceAppointmentTrigger in place.

11.6 CRM backfill and smoke test

Set FS_IsManualInspection__c true on the Work Order, line item, and Service Appointment for a manual booking. The event still publishes, with isManualInspection true. Use a Field Services appointment with BCO Detail.

Step	Expected
Insert a non-manual appointment	No event
Insert a manual inspection	One event, isManualInspection true
Update Is Confirmed to true	One event, isConfirmed true
Scheduled, both times, then assign a resource	One event after the lookup is stamped, with inspector and both times
Change schedule times only	One event with the new times
Change the assigned resource only	One event with the new assignedInspectorId
Arrival window only	No event
Dispatched	One event, status Dispatched
Canceled, with a cancellation reason	One event, status Canceled, cancelledReason set
None while the inspector is still set	No event yet

Step	Expected
Remove the last Assigned Resource at status None	One event, status None, no assignedInspectorId
Is Manual Inspection checked, then schedule or dispatch	One event, isManualInspection true
Subscriber user reads the event	Read succeeds. Create is not required

Sign-off

- [ ] Publisher set assigned to dispatcher, field, booking, and publisher integration users
- [ ] Confirmation process can edit FS_Is_Confirmed__c
- [ ] Subscriber user has Read on FS_InspectionDataUpdate__e and Create only if that user also publishes
- [ ] Subscriber maps InspectionStatusUpdate, status, inspector, schedule, and cancellation fields
- [ ] Is Manual Inspection, Is Confirmed, and Assigned Inspector are on the Field Services layouts
- [ ] Default Setting active; SA and Assigned Resource trigger actions are not bypassed
- [ ] No second MetadataTriggerHandler on Service Appointment or Assigned Resource
- [ ] CRM stamps FS_IsManualInspection__c so the payload identifies manual inspections
- [ ] Smoke tests passed

Not included in this deploy

- FS_InspectionDataUpdate__e object and fields (already in the org).
- Alert Center parent lookup fields and the Field_Services record type.
- TAF object definitions. Only the Service Appointment and Assigned Resource records are deployed.
- A subscriber permission set, named credential, or subscriber credentials.
- Alert Center and FSL managed trigger bodies.